



Operating system and system programming (25CS2104E)

SYSTEM CALLS

Review-I

Under the Esteemed Guidance of

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Contents

- Abstract
- Introduction
- Hardware and Software requirements
- System Architecture
- Conclusion

Abstract

System Calls Project demonstrates how user programs interact with the operating system through system calls.

It focuses on **file handling** operations such as creating, opening, reading, writing, copying, and closing files.

It demonstrates **process management**, including creating, managing, and executing processes and commands.

It handles **keyboard input and command processing**, including input buffers, tokenization, command validation, and syntax error detection.

Overall, the project provides **hands-on knowledge of OS-level programming** and helps understand how the operating system manages files, processes, memory, hardware, and system resources.

Introduction

System Calls Project demonstrates how user applications communicate with the **OS kernel** using system calls.

It covers essential OS services such as **file management, process creation, input/output, and memory management.**

Important system calls like **open(), read(), write(), close(), fork(), and exec()** are implemented using **C and POSIX libraries.**

The project demonstrates how **user commands are received, processed, validated, and executed** by the operating system.

It provides practical knowledge of **user space, system-call interface, kernel space, and hardware resource management.**

Hardware and Software requirements

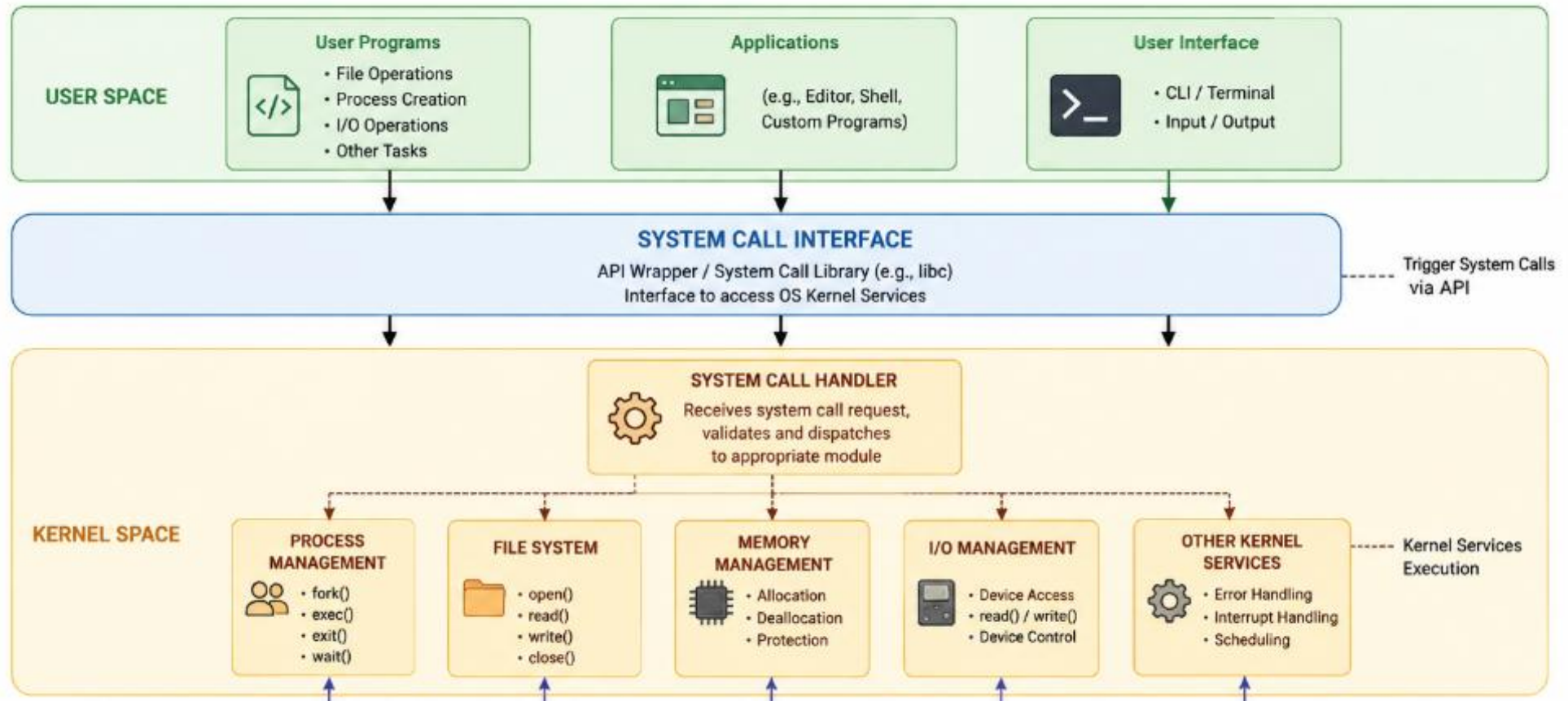
Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB or above
Storage	At least 10 GB free space
Keyboard	Standard USB/Bluetooth keyboard
Display	Standard monitor/display
System	Desktop or Laptop

Software Requirements

Software	Requirement
Operating System	Linux (Ubuntu recommended) / Windows with WSL
Programming Language	C
Compiler	GCC (GNU Compiler Collection)
Code Editor/IDE	Visual Studio Code / Code::Blocks
Terminal	Linux Terminal / VS Code Terminal
Libraries	Standard C libraries and POSIX system-call libraries

System Architecture



Thank you